Service Manual

SERVO-i Ventilator System

MAQUET
GETINGE GROUP

6 Troubleshooting

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6.1 General

Before starting troubleshooting, try to eliminate all possibilities of operational errors. If the malfunction remains, use the troubleshooting guides below as well as the information in chapter Description of functions to locate the faulty part. Perform actions step by step and check that the malfunction is eliminated.

When the fault is corrected, carry out a complete Pre-use check as described in the User's Manual.

The troubleshooting guides below are focused only on technical problems. Information about clinically related problems can be found in the User's Manual.

For functionality enhancement, the latest released System software version is always recommended.

Possible causes to malfunction not mentioned in the following troubleshooting guides are:

- The system has not been correctly assembled after cleaning, maintenance or service.
- Disconnection or bad connection in cable connectors, PC board connectors, and interconnection boards.
- Disconnected or defective gas tubes.

These possible causes to malfunction must always be considered during troubleshooting.

6.2 Pre-use check

The system requests the user to start the automatic Pre-use check at every startup of the system. It is also possible to select the Pre-use check via the Standby menu.

The User's Manual describes how to perform this Pre-use check. The description on the following pages gives more detailed information about the Pre-use check. This information can be used e.g. during troubleshooting of the system.

Some of the recommended actions described below refer to the Field Service System (FSS). The Service key is required to access the FSS. Troubleshooting can of course be performed without access to the FSS, but for some of the recommended actions, the FSS will make troubleshooting faster and easier.

Check if the fault remains after each performed service action. Re-run the complete Pre-use check or run the concerned test using the FSS.

Via the standby menu it is possible to perform a separate Patient circuit test to evaluate circuit leakage and measure circuit compliance. This separate test does not replace the complete Pre-use check.

* Text within brackets refers to the tested subsystem; BRE = Breathing, MON = Monitoring, PAN = Panel.

Test	Test description *	Recommended action if the test fails
During system start-up.	Internal technical tests: • SW check • Reading EEPROM • Checksum EEPROM • Panel button stuck test • Audio test. (BRE + MON + PAN)	1. Restart the system. Do not touch the User interface during system startup. Interfering with the knobs, keys, touch screen, loudspeaker grid, etc, may affect the internal technical tests. 2. Reinstall the System software.
Internal tests	Audio test. (MON + PAN)	If possible, check the Test results log in the More detailed mode (FSS). If the audio test failed: 1. Make sure that the Patient unit main cover and the User interface rear cover are correctly mounted. Otherwise the audio tests may fail. 2. Check in the Test results-log (FSS) if it was the Panel test or the Monitoring test that failed: • If the Monitoring test failed: Replace PC 1772 Monitoring. • If the Panel test failed: Replace the loudspeaker or PC 1777 Panel.
	Alarm output connector test. Performed only if this option is installed. (MON)	If the Alarm output connector test failed: 1. Replace PC 1957 RS232/Alarm output connector (alt. PC 1789 Alarm output connector). 2. Replace PC 1772 Monitoring
	Power failure test. (MON)	If the power test failed: 1. Replace PC 1772 Monitoring.
Barometer test	Checks that the barometric pressure measured by the internal barometer is within 650–1070 hPa. Checks that the measured barometric pressure values differ less than 8 hPa between BRE and MON. (BRE + MON)	Check the Barometric pressure value in the *Status* window. If that value is within 650–1070 hPa: 1. Replace PC 1771 Control 2. Replace PC 1772 Monitoring 3. Replace PC 1784/PC 2024 Expiratory channel 4. Replace the gas modules. Replace one gas module at a time. If that value is outside 650–1070 hPa: 1. Replace PC 1772 Monitoring.

Test	Test description *	Recommended action if the test fails
Gas supply test	The test will start with a 14 s gas flush for Gas type detection: 1. If Air or Heliox is used. 2. That approved Heliox mixture is used. 3. That the Heliox option is enabled (if Heliox gas is detected). Checks that the gas supply pressures measured by the internal gas supply pressure transducers are within: 1. Air and O_2: 200–650 kPa (2.0–6.5 bar). With System version 7.0 (or higher; 200–600 kPa (2.0–6.0 bar). 2. Heliox: 200–350 kPa (2.0–3.5 bar). Checks that the measured supply gas pressure values differs less than 20 mbar between MON and BRE. (BRE + MON)	If the Gas type detection fails: 1. Check that the connected gases are within specified range. 2. If Heliox is used, check that the Heliox option is installed. If the gas supply pressure test fails: 1. Check that the connected gas supply pressure is within the specified range. 2. Start the system in a ventilation mode and check the alarms: • If an Air supply pressure-alarm is activated, replace the Air gas module. Note: If Heliox is used: - Replace the Heliox adapter. - Replace the Air gas module. • If an O_2 supply pressure-alarm is activated, replace the O_2 gas module. 3. Replace PC 1771 Control. 4. Replace PC 1772 Monitoring.

Test	Test description *	Recommended action if the test fails
Internal leakage test	Checks the internal leakage, with test tube connected, using the inspiratory and expiratory pressure transducers. Checks that the leakage at 80 cmH_2O is maximum 10 ml/min. Checks that the measured pressure values differ less than 5 cmH_2O between Insp. and Exp. (BRE)	The Leakage detection, described in chapter Service procedures can be used to locate the leakage. The Leakage detection is a complement to the recommended actions below. If message *Leakage* or *Excessive leakage* appears: 1. Check that the test tube is correctly connected. 2. Check the expiratory cassette: • Check that the cassette is correctly seated in the cassette compartment. • If possible, replace the expiratory cassette and check if the new cassette is accepted by the Pre-use check. • If the new cassette was accepted by the Pre-use check, the fault was located to the cassette. To repair the old cassette, check that the expiratory valve membrane is clean and correctly seated in the cassette. Replace the membrane if required. 3. Check that the pressure transducer tubes/filters are correctly mounted. 4. Check the inspiratory section: • Check that the inspiratory pipe is correctly mounted in the inspiratory section. • Check that the safety valve membrane is clean and correctly seated in the inspiratory pipe. • Check that the safety valve closes properly when the Pre-use check is started (distinct clicking sound from the valve). If the safety valve opens during this test, the opening pressure may not be correctly calibrated (see Safety valve test below). Run the Safety valve test and repeat the complete Pre-use check.

Test	Test description *	Recommended action if the test fails
Internal leakage test (continued)		If message *Pressure Transducer difference > 5 cmH_2O* appears: 1. Check that the pressure transducer tubes and the inspiratory pressure transducer filter are correctly mounted. 2. Check that both PC 1781 Pressure transducer (Insp. and Exp.) are correctly mounted. 3. If the Pressure transducer test also fails (see below), refer to the recommended actions if Pressure transducer test failed. If message *System volume too small* appears: 1. Replace the gas modules. Replace one gas module at a time. Moisture in the gas supply can cause gas module error. If so, this must be addressed so that the error does not recur. If message *System volume too large* appears: 1. Check that the correct test tube is used during the Pre-use check. 2. If the Flow transducer test also fails (see below), replace the gas modules. Replace one gas module at a time. Moisture in the gas supply can cause gas module error. If so, this must be addressed so that the error does not recur. 3. Refer to troubleshooting as described for *Leakage* or *Excessive leakage* above.

Test	Test description *	Recommended action if the test fails
Pressure transducer test	Calibrates and checks the inspiratory and expiratory pressure transducers. The new zero value for the pressure transducers may not differ more than ±6 cmH_2O from factory calibration. With the inspiratory pressure transducers used as a reference, a new gain factor is set for the expiratory pressure transducer. The new gain factor may not differ more than ±5% from factory calibration. During this test, the different subsystems concerned are compared. The difference between the sub-systems must not be more than ±1 cmH_2O at 60 cmH_2O. Expiratory valve coil test. Measures offset and gain in the valve coil. (BRE + MON)	Check the Test results log in the More detailed mode (FSS). If Pressure transducer test failed: 1. Cancel the Pressure transducer test and continue with the remaining tests (this will calibrate the safety valve). Restart the Pre-use check. 2. Check the expiratory cassette: • If possible, replace the expiratory cassette and check if the new cassette is accepted by the Pre-use check. • If the new cassette was accepted by the Pre-use check, the fault was located to the cassette. The fail with the old cassette may in this case be due to water collected at the pressure transducer filter inside the cassette. Dry the cassette properly. 3. Check/replace PC 1781 Pressure transducer (Insp. and Exp.). To locate the faulty pressure transducer, replace one transducer at a time. 4. Replace PC 1771 Control. 5. Replace PC 1772 Monitoring. If Expiration valve test failed: 1. Check the expiratory cassette: • If possible, replace the expiratory cassette and check if the new cassette is accepted by the Pre-use check. • If the new cassette was accepted by the Pre-use check, the fault was located to the cassette. To repair the old cassette, check that the expiratory valve membrane is clean and correctly seated in the cassette. Replace the membrane if required. 2. Replace the Expiratory valve coil.
Safety valve test	Checks and if necessary adjusts the opening pressure for the safety valve to 117 ±3 cmH_2O. Checks the hardware signals related to the safety valve functions. (BRE + MON)	1. Check the inspiratory section: • Check that the inspiratory pipe is correctly mounted in inspiratory section. • Check that the safety valve membrane is clean and correctly seated in the inspiratory pipe. 2. Replace the safety valve pull magnet. 3. Replace PC 1784/PC 2024 Expiratory channel. 4. Replace PC 1772 Monitoring.

Test	Test description *	Recommended action if the test fails
O_2 cell/sensor test	Calibrates and checks the O_2 sensor/cell at 21% O_2 and 100% O_2. Checks if the O_2 cell is worn out. As different gas mixtures are used during this test, calibration and check of O_2 sensor/cell will not be performed if one gas is missing. The O_2 sensor test requires 21% O_2 and 100% O_2. If other gas mixtures are used (e.g. O_2 supply delivers 96% O_2 instead of 100%) the O_2 cell/sensor test may fail. (BRE + MON)	1. Check that the connected gas supply pressure (Air and O_2) is within the specified range. 2. Replace the O_2 sensor/cell. 3. Replace the gas modules. Replace one gas module at a time. 4. Replace PC 1771 Control. 5. Replace PC 1772 Monitoring.
Flow transducer test	Checks the inspiratory flow transducer. Calibrates and checks the expiratory flow transducer. Calibrates at 60% O_2 and checks at 100% and 21% O_2. As different gas mixtures are used during this test, calibration of the expiratory flow transducer will only be performed if both gases are connected. The check using the connected gas, (100% alt. 21% O_2) will however be performed. The Flow transducer test will pass if the result of this check corresponds to the old calibration factor from a previous Pre-use check. The same expiratory cassette must be used. The new calibration factor for the expiratory flow transducer may not differ more than -10% to +15% from factory calibration. During this test, the different subsystems concerned are compared. The difference between the subsystems must not be more than ±0.3 l/min. (BRE + MON)	1. Check that the connected gas supply pressure (Air and O_2) is within the specified range. 2. Check the expiratory cassette: • Check that the cassette is correctly seated in the cassette compartment. • If possible, replace the expiratory cassette and check if the new cassette is accepted by the Pre-use check. • If the new cassette was accepted by the Pre-use check, the fault was located to the cassette. The fail with the old cassette may in this case be due to water collected inside the cassette. Dry the cassette properly. 3. Replace the gas modules. Replace one gas module at a time. 4. Replace PC 1785 Expiratory channel connector. 5. Replace PC 1771 Control. 6. Replace PC 1772 Monitoring. 7. Replace PC 1784/PC 2024 Expiratory channel.
Battery switch test	Checks that the power supply switches to battery when mains power is disconnected. Checks that the power supply switches back to mains power when reconnected. This test will not be performed if: • Less than 10 min. backup time remains in the connected Battery module(s). • No Battery module is connected. (MON)	1. Check that the total remaining time for the Battery module(s) is >10 min. If not, allow the battery to charge and repeat the test. 2. Replace PC 1775 Plug-and-play back-plane. 3. Replace the Battery module(s).

Test	Test description *	Recommended action if the test fails
Patient circuit test	Checks the patient circuit leakage, compliance and resistance, with patient tubing connected, using the inspiratory and expiratory pressure transducers. Allowed leakage: 80 ml/min at 50 cmH_2O. Will allow the system to calculate a compensation for circuit compliance (if the leakage requirements are met). Checks the patient circuit resistance, with patient tubing connected, using the inspiratory and expiratory pressure transducers. (BRE)	If the internal leakage test (see above) has passed, the leakage is to be located to the patient circuit. Check for leakage or replace the patient circuit.
Y sensor test	Checks the pressure and flow measurement of the Y sensor. This test is only performed if a Y sensor and a Y sensor module is connected to the system.	1. If the patient circuit test (see above) has passed, check that there is no leakage in the Y sensor and that there are no obstructions in the Y sensor tubings. 2. Replace the Y sensor. 3. Replace the Y sensor module.
Alarm state test	Checks that no Technical error alarms are active during the Pre-use check. (MON)	Refer to section regarding Technical error codes and messages for further information.
External alarm system test	If the option Alarm output connector is enabled, the user can test the external alarm system. The external alarm output signal is activated and the user must verify the external alarm.	1. Check the external alarm system. 2. Replace PC 1957 RS232/Alarm output connector (alt. PC 1789 Alarm output connector). The result of the test does not affect the outcome of the Pre-use check.
Separate Patient circuit test	This test separately performs the Patient circuit test as described above. If a Y sensor and a Y sensor module are connected to the system, the Y sensor test is also performed (as described above).	See above.

6.3 Technical error codes and messages

The table below shows recommended actions in case of Technical errors.

Some of the error codes are intended only for R&D, not for field service. If so, the text N/A is stated in the Recommended action column.

6.3.1 Monitoring

Error code	Error message / Possible cause	Recommended action
1	Power error. -12 V too low, < -13.2 V.	1. Replace PC 1778 DC/DC & Standard connectors.
2	Power error. -12 V too high, > -10.8 V.	1. Check status of external battery (if connected). 2. Replace PC 1778 DC/DC & Standard connectors. 3. Replace PC 1775 Plug-and-play back-plane.
3	Power error. +12 V too low, < +10.8 V.	1. Check status of external battery (if connected). 2. Replace PC 1778 DC/DC & Standard connectors. 3. Replace PC 1775 Plug-and-play back-plane.
4	Power error. +12 V too high, > +13.2 V.	1. Replace PC 1778 DC/DC & Standard connectors.
5	Power error. +24 V too low, < 21.5 V.	1. Replace PC 1778 DC/DC & Standard connectors. 2. Replace the gas modules. Replace one gas module at a time and check that this technical error code will not appear.
6	Power error. +24 V too high, > 26.5 V.	1. Replace PC 1778 DC/DC & Standard connectors.
7	Software error. Patient category mismatch between Breathing and Monitoring subsystems.	1. Replace PC 1771 Control. 2. Replace PC 1772 Monitoring. 3. Replace the gas modules. Replace one gas module at a time and check that this technical error code will not appear.
8	Insp. pause hold time exceeded. The maximum *Insp. pause hold time* 30 s exceeded. The system has not returned to ventilation after *Insp. pause hold time*.	1. Replace PC 1771 Control.
9	Exp. pause hold time exceeded. The maximum *Exp. pause hold time* 30 s exceeded. The system has not returned to ventilation after *Exp. pause hold time*.	1. Replace PC 1771 Control.

Error code	**Error message / Possible cause**	**Recommended action**
10	Ventilation stopped. The VALVES_DISABLED signal has stopped power supply to gas modules and safety valve.	1. Replace PC 1784/PC 2024 Expiratory channel. 2. Replace PC 1771 Control. 3. Replace PC 1778 DC/DC & Standard connectors. 4. Replace PC 1772 Monitoring.
11	Safety valve open. The Safety valve is detected as open without fulfilled opening conditions.	1. Check inspiratory channel. 2. Replace safety valve pull magnet. 3. Replace PC 1784/PC 2024 Expiratory channel. **Note:** If the Patient unit is placed too close to the MR scanner, this error may be activated.
12	Ventilation stopped. PC 1771 Control error. Communication with PC 1772 Monitoring lost due to software or hardware failure.	1. Replace PC 1771 Control.
13	Ventilation stopped. Software conflict. Two PC 1771 Control detected on the CAN bus.	N/A
15	Ventilation will most probably continue, but communication with PC 1777 will be lost. Software conflict. Two PC 1777 Panel detected on the CAN bus.	N/A
16	Ventilation error. PC 1784/PC 2024 Expiratory channel failure.	1. Replace PC 1784/PC 2024 Expiratory channel.
17	Ventilation stopped. Software conflict. Two PC 1784/PC 2024 Expiratory channel detected on the CAN bus.	N/A
22	Backup audible alarm error. Backup alarm buzzer error.	1. Replace PC 1772 Monitoring.
24	Backup audible alarm or Remote alarm error. Error indicated in the backup alarm or remote alarm capacitor on PC 1772 Monitoring.	1. Make sure that System SW version 3.02.02 (or higher) is installed. 2. Replace PC 1772 Monitoring

Error code	Error message / Possible cause	Recommended action
25	Communication error. Communication error between ID PROMs/EEPOTs and Monitoring subsystem. The ID # is followed by an Error # in the Technical alarm log. For Error 6, report to MCC HSC / Maquet for further information. For all other Error #, replace the part identified by the ID #. Error # 1. CHK_SUM_ERR 2. NR_OF_BYTES_IN_CHKSUM_ERR 3. CHKSUM_FILL_BYTE_ERR 4. WRITABLE_RECORDS_FAILURE 5. WRITABLE_RECORDS_DIFFER 6. COMM_ERR 7. VALIDATION_ERR 8. WCR_VALIDATION_ERR 9. WRITE_NOT_PERMITTED 10. OTHER_ERR	Depending on ID # stated in the Technical alarm log. If repeated, replace parts as follows: 0: Main back-plane. 1: PC 1772 Monitoring. 2: PC 1775 Plug-and-play back-plane. 3: PC 1778 DC/DC & Standard connectors. 4: PC 1781 Inspiratory pressure transducer. 5: PC 1781 Expiratory pressure transducer. 6: O_2 sensor/cell or O_2 sensor/cell cable or PC 1772 Monitoring. 32: PC 1784/PC 2024 Expiratory channel or PC 1772 Monitoring. 33: Air gas module or PC 1772 Monitoring. 34: O_2 gas module or PC 1772 Monitoring. 35: PC 1772 Monitoring. 256: PC 1771 Control. 512: PC 1777 Panel. 784: N/A
27	Backup audible alarm error. Backup alarm buzzer error.	1. Replace PC 1772 Monitoring. **Note:** If the Patient unit front cover is removed, this error may be activated.
28	Audible alarm error. Loudspeaker error.	1. Replace the loudspeaker. 2. Replace PC 1777 Panel. **Note:** If the User interface rear cover is removed, this error may be activated.
29	Memory backup battery depleted. The Memory backup battery on PC 1772 Monitoring is depleted.	1. Replace Memory backup battery on PC 1772 Monitoring.
32	Software version conflict. Conflict between the software versions installed.	N/A
33	Ventilation disabled. Communication error or PC 1771 Control error during startup.	1. Replace PC 1771 Control.

Error code	Error message / Possible cause	Recommended action
34	User interface communication error. Communication error between PC 1777 Panel and PC 1772 Monitoring during startup.	1. Check the control cable. 2. Replace PC 1778 DC/DC & Standard connectors. 3. Replace PC 1777 Panel. **Note:** This error indicates communication error between PC 1777 and PC 1772 and the error code will thus not be shown on the display (but will be logged).
35	Ventilation disabled. Communication error or PC 1784/PC 2024 Expiratory channel failure during startup. May appear at the first startup after a software installation (restart required).	1. Replace PC 1771 Control (if logged after error code 40001). 2. Replace PC 1784/PC 2024 Expiratory channel.
37	Software error. Patient category mismatch between Breathing and Monitoring subsystems.	N/A
38	Barometric pressure too high. Barometric pressure above allowed limit or barometer error.	1. Adjust barometer (FSS required). 2. Replace PC 1772 Monitoring.
39	Barometric pressure too low. Barometric pressure below allowed limit or barometer error.	1. Adjust barometer (FSS required). 2. Replace PC 1772 Monitoring. **Note:** This alarm will be activated if the ambient pressure is below 650 hPa, e.g. on a high altitude.
40	Measured value invalid. Invalid measured metric detected (Ppeak, MVe or RR). The invalid metric is not displayed.	1. Restart the system. 2. Check that System SW version 4.00.04 (or higher) is installed. Update software if required.
41	Real time clock error. Internal processor real time clock error.	1. Replace PC 1772 Monitoring.
42	Checksum error. Software checksum error during startup.	1. Restart the system. 2. Replace Memory backup battery on PC 1772 if together with Error code 29. 3. Replace PC 1772 Monitoring.
43	Battery information error. Battery module status information error.	1. Replace the Battery module(s). 2. Replace PC 1775 Plug-and-play back-plane. 3. Replace PC 1772 Monitoring.

Error code	Error message / Possible cause	Recommended action
44	Alarm limit error. Alarm limit settings (with checksum) are stored. If the checksum is incorrect, this Technical error alarm is activated.	1. Restart the system. 2. Replace Memory backup battery on PC 1772 if together with error code 29. 3. Replace PC 1772 Monitoring.
45	Unexpected shutdown. Unexpected shutdown. As the system has shut down, this Technical Error is only visible in the Technical alarm log.	1. Restart the system. 2. Replace PC 1772 Monitoring.
46	Alarm output connector error. Alarm output connector error or communication error with PC 1772.	1. Replace PC 1957 RS232/Alarm output connector (alt. PC 1789 Alarm output connector). 2. Replace PC 1772 Monitoring.
48	Pre-oxygen time exceeded. The maximum *Pre-oxygen time* during Suction Support exceeded. The system has not returned to ventilation using the previous settings after the *Pre-oxygen time*.	1. Replace PC 1771 Control.
49	Disconnect time exceeded. The maximum *Disconnect time* during Suction Support exceeded. The system has not returned to ventilation using the previous settings after the *Disconnect time*.	1. Replace PC 1771 Control.
50	Communication error. Communication error between ID PROM on PC 1770 Main back-plane and Monitoring subsystem. System ID, configuration, operating time, etc, not available.	1. Replace PC 1772 Monitoring. 2. Replace PC 1770 Main back-plane.
51	On/Off switch error. The On/Off switch is neither in On or Off position during more than 3 seconds. Applicable only on systems with PC 1772 Rev 18 (or higher) and System SW version 3.02.01 (or higher).	1. Check/replace Control cable. 2. Replace PC 1777 Panel.
52	O_2 sensor cannot be used with Heliox.	1. Replace the O_2 sensor with an O_2 cell.
53	PC 1784 Expiratory channel version conflict.	1. Replace PC 1784 Expiratory channel. PC 1784 must be of Revision 16 (or higher) when used with Heliox. Also note that PC 1785 Expiratory channel connector of Revision 03 (or higher) is a prerequisite for Heliox.
54	System ID conflict. Conflict during startup between information stored on PC 1770 Main back-plane and PC 1772 Monitoring.	1. Restart the system. 2. Replace Memory backup battery on PC 1772 if together with error code 29. 3. Replace PC 1772 Monitoring.

6.3.2 Breathing

Error code	Error message / Possible cause	Recommended action
10001	Lithium battery depleted. The Memory backup battery on PC 1771 Control is depleted.	1. Replace battery on PC 1771 Control.
10002	Communication error. Communication error with ID PROM on PC 1771 Control.	1. Replace PC 1771 Control.
10003	Ventilation stopped. Error in the internal memory detected. This technical alarm is common after a software installation. Restart the system and check that no technical alarms are activated. If error code 10003 remains, this indicates a hardware error.	1. Restart the system. 2. Replace PC 1771 Control.

6.3.3 Panel

Error code	Error message / Possible cause	Recommended action
20001	Communication error. Communication error with ID PROM on PC 1777 Panel.	1. Replace PC 1777 Panel.
20002	Backlight error. The backlight power consumption is measured. If the power consumption is outside specified range during >20 consecutive readings, this technical error is activated. This error message may also be logged immediately before the *No battery capacity* alarm when the system is running in battery mode.	If the display is lit (and the error code is shown on the display): 1. Replace PC 1777 Panel. If the display is not lit (the display is dark): 1. Replace backlight lamps. 2. Replace PC board Backlight Inverter. **Notes:** • If one of the lamps is broken, the other lamp will automatically be switched off. Thus, with a failure on a backlight lamp, or likely also on PC board Backlight inverter, the User interface display will become dark. This error code will in such case not be possible to see (but will be logged). • If the User interface is placed too close to the MR scanner, the PC board Backlight inverter may become defective.
20003	Membrane button error. Membrane buttons are checked during startup. If any button is detected as pressed, this technical error is activated.	1. Restart the system. Do not touch the User interface during system startup. Interfering with the knobs, keys, touch screen, loudspeaker grid, etc, may affect the internal technical tests. 2. Check the User interface membrane buttons (FSS). 3. Replace the Touch screen. 4. Replace PC 1777 Panel.
20004	Alarm sound level too low. The loudspeaker is checked using a microphone during startup: • On PC 1777A-C; one beep (silent before and after). • On PC 1777D (or higher); three beeps (silent between).	1. Restart the system. Do not touch the User interface during system startup. Interfering with the knobs, keys, touch screen, loudspeaker grid, etc, may affect the internal technical tests. 2. Replace the loudspeaker. 3. Replace PC 1777 Panel.
20005	Checksum error. Software checksum failure during startup.	1. Replace PC 1777 Panel.

6.3.4 Expiratory Flow Meter

Error code	Error message / Possible cause	Recommended action
For systems with PC 1784 Expiratory channel		
40001	Exp. flow meter error. See information logged together with error code 40001 in the Service log / Technical alarms.	Technical alarm *Expiration flow meter PC 1784* and *Exp. cassette EEPROM read/write error*. 1. Replace PC 1784 Expiratory channel. Technical alarm *Expiration flow meter PC 1784* and *Exp. cassette EEPROM checksum error:* 1. Replace Expiratory cassette. Technical alarm *Expiration flow meter PC 1784* and *PC 1784 thermistor failure*: 1. Check fan operation for proper cooling. 2. Replace PC 1784 Expiratory channel. Technical alarm *Expiration flow meter PC 1784* and *PC 1784 60 V overrange*: 1. Replace PC 1784 Expiratory channel. Technical alarm *Expiration flow meter PC 1784* and *PC 1784 60 V underrange*: 1. Replace PC 1771 Control. 2. Replace PC 1778 DC/DC & Standard connectors. 3. Replace PC 1784 Expiratory channel.
For systems with PC 2024 Expiratory channel		
40001	Internal communication error on PC 2024.	1. Replace PC 2024 Expiratory channel.
40002	Temperature sensor error on PC 2024. Note that this error may occur if the system is running outside operating temperature range.	1. Replace PC 2024 Expiratory channel.
40003	Power error. +60 V supply voltage to the ultrasonic transducers in the Expiratory cassette too high, > +75 V.	1. Replace PC 2024 Expiratory channel.
40004	Power error. +60 V supply voltage to the ultrasonic transducers in the Expiratory cassette too low, < +45 V.	1. Replace PC 2024 Expiratory channel. 2. Replace PC 1778 DC/DC & Standard connectors.
40005	Power error. +1.8 V is more than 15 % outside nominal value.	1. Replace PC 2024 Expiratory channel.

Error code	Error message / Possible cause	Recommended action
40006	Power error. +1.2 V is more than 15 % outside nominal value.	1. Replace PC 2024 Expiratory channel.
40007	Power error. +5 V is more than 15 % outside nominal value.	1. Replace PC 2024 Expiratory channel. 2. Replace PC 1778 DC/DC & Standard connectors.
40008	Power error. -5 V is more than 15 % outside nominal value.	1. Replace PC 2024 Expiratory channel.
40009	Power error. +2.5 V is more than 15 % outside nominal value.	1. Replace PC 2024 Expiratory channel.
40010	Internal temperature too high, > 77 °C. The temperature is too high or temperature sensor error on PC 2024.	1. Ensure proper cooling: - Check fan operation - Check the operating temperature - Check the fan filter in the Patient unit. 2. Replace PC 2024 Expiratory channel.
40011	Multiple CPU restarts on PC 2024.	1. Restart the system. 2. Replace PC 2024 Expiratory channel.

6.3.5 Other Technical error alarms

Error message / Possible cause	Recommended action
Restart ventilator Communication error between PC 1777 Panel and PC 1772 Monitoring.	1. Restart the system and perform a Pre-use check. 2. Check the Control cable that connects the Patient unit with the User interface. 3. Replace PC 1778 DC/DC & Standard connectors. 4. Replace PC 1772 Monitoring. 5. Replace PC 1777 Panel.
Technical error in Expiratory cassette Technical error in Expiratory cassette or in the communication with the cassette.	1. Replace the expiratory cassette. 2. Replace PC 1785 Expiratory channel connector. 3. Replace PC 1784 Expiratory channel. Check the *Service log / Technical alarms*. If *Expiration flow meter Exp. cassette power failure* is logged together with error code *Technical error in Expiratory cassette*: 1. Replace PC 1771 Control. 2. Replace PC 1778 DC/DC & Standard connectors.
CO_2 module error Hardware error in the CO_2 analyzer module.	1. Unplug and reinsert the module. 2. Replace the module. There are no spare parts available for the CO_2 analyzer module. In case of malfunction, the module must be replaced.

Error message / Possible cause	Recommended action
CO_2 sensor error Hardware error in CO_2 capnostat sensor. The values in the capnostat memory failed the internal test.	1. Unplug and reinsert the capnostat sensor. 2. Calibrate the capnostat sensor. 3. Replace the capnostat sensor. There are no spare parts available for the capnostat sensor. In case of malfunction, the sensor must be replaced.
Nebulizer hardware error Technical problem with nebulizer hardware. Temperature in the nebulizer patient unit too high. Technical problem with connection cable.	1. Check buffer liquid level. 2. Restart the nebulizer. 3. Replace the nebulizer patient unit. 4. Replace connection cable. 5. Check internal connection cable. 6. Replace PC 1771 Control.
Edi module error Hardware error in the Edi module.	1. Unplug and re-insert the Edi module. 2. Replace the Edi module. There are no spare parts available for the Edi module. In case of malfunction, the module must be replaced.
Y sensor module error Hardware error in the Y sensor measuring module.	1. Unplug and reinsert the module. 2. Replace the module. There are no spare parts available for the Y sensor module. In case of malfunction, the module must be replaced.